INCH-POUND

MIL-PRF-9050H
12 June 2020
SUPERSEDING
MIL-PRF-9050G
25 AUGUST 2009

PERFORMANCE SPECIFICATION

VALVES, OXYGEN, PRESSURE RELIEF, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

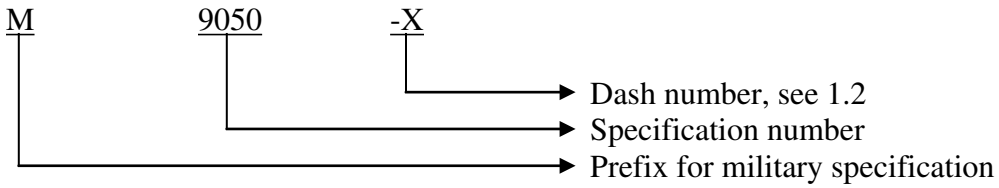
1. SCOPE

1.1 Scope. This specification covers aircraft pressure relief valves for 70 and 300 psig rated liquid oxygen (LOX) converters.

1.2 Classification. The valves will be of the following types and Government designations, as specified (see 6.2):

Type	PIN	Operating Range
I	M9050-1	80-100 psig
II	M9050-2	100-120 psig
III	M9050-3	330-380 psig
IV	M9050-4	380-430 psig
V	M9050-5	120-140 psig

1.3 Part or Identifying Number (PIN). The PIN to be used for valve acquired to this specification is created as follows:



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	-	Nitrogen, Technical
A-A-59155	-	Nitrogen, High Purity, Special Purpose

(Copies of these documents are available online at https://assist.dla.mil)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27210	Oxygen, Aviation's Breathing, Liquid and Gas

(Copies of these documents are available online at https://assist.dla.mil)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(ASQ documents may be obtained at http://asq.org or addressed to American Society for Quality P.O. Box 3005, Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, Wisconsin, 53203, USA)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	-	Salt Spray (Fog) Apparatus, Operating
ASTM D1149	-	Rubber Deterioration-Surface Ozone Cracking In A Chamber

(ASTM documents may be obtained at www.astm.org or addressed to American Society for Testing and Material, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 6.3).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.3 Materials. All materials shall be corrosion resistant or suitably treated to resist corrosion due to electrolytic decomposition, salt fog, and any other atmospheric condition that may be encountered during operational use or storage. All material used shall be capable of operating safely in a pure oxygen and LOX environment. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Ozone resistance. Elastomer materials shall be ozone resistant.

3.3.2 Age. Elastomers, except silicone, shall be no more than 12 months old from the cure date upon delivery.

3.4 Interface.

3.4.1 Dimensions. The dimensions of the valve shall conform to figure 1.

3.4.2 Threads. All pipe threads shall be in accordance with SAE-AS71051.

3.5 Performance.

3.5.1 Weight. The weight of the valve shall not exceed 0.25 pound.

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3.5.2 Leakage. The valve shall not leak at a rate greater than 0.01 liter per minute (LPM) when subjected to an internal pressure at the minimum operating pressure of the valve (see 1.2) and after establishing a 15 LPM flow through the valve.

3.5.3 Flow capacity. The valve shall have a minimum gas flow rate of 100 LPM at the maximum operating pressure (see 1.2).

3.5.4 Sustained flow. The valve shall withstand a gas flow rate of 100 LPM for 1 hour without chatter (see 6.6).

3.5.5 Endurance. The valve shall withstand 5,000 endurance cycles of operation. For each endurance cycle, the valve shall be pressurized from below the minimum operating pressure to the maximum operating pressure and back to below the minimum operating pressure (see 1.2). The valve shall seat each time at the minimum operating pressure.

3.5.6 Overpressure cycling. The valve shall be pressurized from below the minimum operating pressure to a pressure of 50 psig above the maximum operating pressure and back to below the minimum operating pressure (see 1.2). This shall constitute one overpressure cycle. The valve shall not leak at a rate greater than 0.01 LPM after the 5th, 15th, and 25th pressurization cycles.

3.5.7 Orientation. The valve shall operate in any orientation.

3.5.8 Tamper-proof relief pressure setting mechanism. The valve shall be equipped with a relief pressure setting mechanism which shall be tamper-proof.

3.5.9 Inlet filter. The inlet of the valve shall have a 75 micron filter or screen to prevent foreign material from entering the valve.

3.6 Environmental conditions.

3.6.1 Temperature. The valve shall operate between LOX exposure temperatures and 265° ± 5°F.

3.6.2 Vibration. The valve shall withstand resonant frequencies as shown in table I (see 6.9) and cycling frequencies between 5 and 500 Hz at an applied double amplitude of 0.036 inch or an applied acceleration of ±10g.

TABLE I. Vibration data

Frequency (Hz)	Displacement in double amplitude (inch) and acceleration (g), if applicable
5-10	0.080
10-10.9	±0.41g (0.080 to 0.067)
10.9-16.2	0.067
16.2-18.6	±0.9g (0.067 to 0.051)

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TABLE I. Vibration data (Continued)

18.6-23.2	0.051
23.2-27.6	±1.4g (0.051 to 0.036)
27.6-73.7	0.036
73.7-500	±10.0g (0.036 to 0.001)

3.7 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 milligrams per square foot (mg/ft^2)(see 6.4).

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Item identification. The valve shall be marked for identification with the following information (see 6.2):

a. Item name.
b. Valve type.
c. Manufacturer's name.
d. Manufacturer's CAGE code.
e. Manufacturer's part number (PN).
f. Part or identification number (PIN) (see 1.3).
g. Elastomer cure date (if applicable).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification (see 4.2).
b. Conformance (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on a minimum of two valves of each type, and shall include all the tests described in 4.6.

4.3 Conformance inspection. Conformance inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual tests. Each valve shall be subjected to the following tests:

a. Examination (see 4.6.1)
b. Leakage (see 4.6.4)
c. Flow capacity (see 4.6.5)

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4.3.2 Sample tests. After completion of the tests in 4.3.1, a sample of valves shall be selected in accordance with ASQ/ANSI Z1.4 at a normal inspection level unless modified in the procurement document (see 6.2), and be subjected to the following tests:

a. Dimensions (see 4.6.2).
b. Sustained flow (see 4.6.6).
c. Endurance (see 4.6.10).
d. Orientation (see 4.6.12).

4.4 Test conditions.

4.4.1 Test medium. The test medium used in qualification testing shall be gaseous or liquid oxygen conforming to MIL-PRF-27210, types I and II. Unless otherwise specified in the individual test description, the test medium used in conformance testing shall be gaseous or liquid oxygen conforming to MIL-PRF-27210, types I and II or gaseous nitrogen conforming to A-A-59503 and A-A-59155, type I, grade B, class 1 (see 6.5).

4.4.2 Temperature and pressure. Unless otherwise specified in the individual test description, tests shall be conducted at ambient temperature and barometric pressure.

4.5 Requirements cross-reference matrix. Table II provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE II. Requirements cross-reference matrix

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.5	4.6.10, 4.6.15
3.3	4.6.1, 4.6.8, 4.6.13	3.5.6	4.6.11
3.3.1	4.6.14	3.5.7	4.6.12
3.3.2	4.6.1	3.5.8	4.6.1
3.4.1	4.6.2	3.5.9	4.6.1
3.4.2	4.6.2	3.6.1	4.6.7, 4.6.8
3.5.1	4.6.1	3.6.2	4.6.9
3.5.2	4.6.4	3.7	4.6.3
3.5.3	4.6.5	3.8	4.6.1
3.5.4	4.6.6	3.9	4.6.1

4.6 Tests.

4.6.1 Examination. The valve shall be inspected to determine conformance to materials, age, weight, tamper-proof relief pressure setting mechanism, inlet filter, interchangeability, and item identification.

4.6.2 Dimensions. The valve shall be inspected to determine conformance to dimensions and threads (see figure 1).

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4.6.3 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 mg/ft² (see 6.4). Cleanliness of the surfaces shall be demonstrated by industrially accepted methods and these cleaning and verification methods shall be identified (see 6.2 and 6.4).

4.6.4 Leakage. The inlet of the valve shall be pressurized to the maximum operating pressure (see 1.2) and then decreased to the minimum operating pressure. While the pressure is maintained at the minimum operating pressure, the valve leakage shall not exceed 0.01 LPM. Pressure shall then be applied to establish a gas flow of 15 ± 10 LPM through the valve. Then the inlet pressurization shall be reduced to minimum operating pressure allowing the internal pressure to decrease until the valve seats. The valve leakage after seating shall not exceed 0.01 LPM.

4.6.5 Flow capacity. The inlet of the valve shall be pressurized to a pressure not to exceed the maximum operating pressure (see 1.2) of the valve being tested. While maintaining the inlet at the maximum operating pressure, the valve shall have a minimum flow of 100 LPM.

4.6.6 Sustained flow. The valve shall be pressurized with oxygen to maintain a flow of 100 LPM through the valve for 1 hour. The valve shall not chatter (see 6.6). After the 1 hour flow period, the valve shall be subjected to the leakage and capacity tests in 4.6.4 and 4.6.5.

4.6.7 Temperature exposure. The valve shall be exposed for 4 hours at 265° ± 5°F. After exposure, and while at 265° ± 5°F, the valve shall be subjected to the leak and capacity tests in 4.6.4 and 4.6.5 (see 6.7).

4.6.8 LOX exposure. The inlet of the valve shall be pressurized with LOX to the maximum operating pressure of the valve. After maintaining liquid flow for a 5 minute period at maximum pressure, the pressure shall be reduced to the minimum operating pressure of the valve. While maintaining the minimum operating pressure for one minute, there shall be no leakage of liquid such as droplets, mist, or vapor (see 6.8). This LOX exposure cycle shall be repeated three times. With the minimum operating pressure remaining in the valve, the measured leakage rate of gaseous oxygen shall not exceed 0.1 LPM after 5 minutes and 0.01 LPM after 10 minutes.

4.6.9 Vibration. The vibration test shall be conducted with the valve pressurized with gas below minimum operating pressure. Tests shall be conducted under both resonance and cycling conditions as specified in paragraphs 4.6.9.1 and 4.6.9.2, respectively. Time consumed performing the resonance test plus time consumed performing the cycling test shall equal 3 hours. The valve shall then be subjected to leakage and capacity tests in 4.6.4 and 4.6.5.

4.6.9.1 Resonance test. Resonant frequencies of the valve shall be determined by varying the frequency, at the applied displacement and acceleration, through the frequency range as shown in table I. Resonant frequency surveys shall be conducted for each of three mutually perpendicular axes of the valve. The valve shall be vibrated at any indicated resonant frequencies for the time shown in table III at the applied displacement shown in table I (see 6.9). Vibration shall be applied along each of the three mutually perpendicular axes. If more than one resonant frequency

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is encountered when vibration is applied along any one axis, vibration at each of these resonant frequencies shall be sustained for the time shown in the applicable portion of the vibration test schedule. If more than four resonant frequencies are encountered when vibration is applied along one axis, the four most severe resonant frequencies shall be chosen for test.

4.6.9.2 Cycling test. The valve shall be vibrated in accordance with the vibration test schedule in table III. The frequency shall be varied between 5 and 500 Hz and then returned to 5 Hz in 15 minute intervals at an applied double amplitude of 0.036 inch or an applied acceleration of ±10.0g, whichever is the lower g value. The rate of change of frequency shall be logarithmic.

TABLE III. Vibration test schedule

Number of resonances	0	1	2	3	4
Total resonance test vibration time (min) (see note)	-	30	60	90	120
Cycling test time (hr)	3	2.5	2	1.5	1
Note: 30 minutes at each resonance					

4.6.10 Endurance. The inlet pressure of the valve shall be cycled from below the minimum operating pressure to the maximum operating pressure of the valve and back to below the minimum operating pressure. The valve shall seat at the end of each cycle when the minimum operating pressure is reached. This pressure endurance cycling shall be repeated 5,000 times. The valve shall then be subjected to the leak and capacity tests in 4.6.4 and 4.6.5.

4.6.11 Overpressure cycling. The valve shall be subjected to 25 overpressure cycles. The inlet pressure of the valve shall be increased to a minimum of 50 psig above the maximum operating pressure and maintained for at least 10 seconds. The inlet pressure shall then be turned off, allowing the internal pressure to decrease to the minimum operating pressure at which point the valve seats. This shall constitute one overpressure cycle. After the 5th, 15th, and 25th cycles, the leakage rate after the seating shall not exceed 0.01 LPM.

4.6.12 Orientation. The valve shall be subjected to the leak and capacity tests in 4.6.4 and 4.6.5 while the valve is mounted in each of the following orientations:

a. Centerline of the inlet horizontal.
b. Centerline of the inlet vertical and the inlet down.
c. Centerline of the inlet vertical and the inlet up.

4.6.13 Salt fog. Unless waived (see 6.2), the valve, with inlet and outlet plugged, shall be subjected to the salt fog test in ASTM B117 for 50 hours. There shall be no visible damage to the valve. The valve shall be subjected to the leakage and capacity tests in 4.6.4 and 4.6.5.

4.6.14 Ozone resistance. Unless waived (see 6.2), samples of elastomer and seating materials shall be tested in accordance with ASTM D1149. The test samples shall be elongated 20%, placed in an ozone-free atmosphere for 24 hours, and then subjected to an ozone environment.

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The temperature shall be 100° ± 2°F, an ozone concentration of 120 ± 10 parts per million by volume, and the air velocity across the sample shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test samples shall be examined under 10x magnification for evidence of damage such as blooming, checking, or cracking.

4.6.15 Disassembly and examination. At the conclusion of all the tests, each valve shall be disassembled and examined. The valve shall show no evidence of deterioration or wear.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The pressure relief valves covered by this specification are intended for use with both the 70 and 300 psig rated liquid oxygen converters and other aircraft system applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Item name and valve type (see 1.2 and 3.9a and b).
c. If the conformance inspection sampling inspection level is other than normal (tightened or reduced) (see 4.3.2).
d. The requirement for the vendor to identify proposed cleaning and verification methods (see 4.6.3).
e. Whether the salt fog and ozone tests are waived and material certification is acceptable (see 4.6.13 and 4.6.14).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 9050 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be

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6.4 Cleaning. See MIL-STD-1330 and SAE-ARP1176 for guidance on proven cleaning methods and verifications. Visible inspections are typically conducted using white light and ultraviolet light to detect particulates and some types of hydrocarbons. The non-volatile residue (NVR) test in SAE-ARP1176 may be used to baseline the hydrocarbon verification at an acceptable contamination level; however, other cleaning and verification methods that do not contain class I and II ODC solvents should be used for production.

6.5 Test medium. When gaseous nitrogen is used as a test medium the appropriate density corrections will be made in determining that the test requirements are met.

6.6 Chatter. Chatter is the sound produced by the rapid and unstable movement of internal components of the valve during operation. Chatter will cause damage to the valve and may create a fire hazard.

6.7 Temperature exposure guidance. A decrease of not more than 5 psig below the minimum pressure has been allowed during testing.

6.8 LOX exposure guidance. Actual leakage should be distinguished from vapor formed by moisture from the surrounding air.

6.9 Vibration. The relationship between frequency, displacement, and acceleration is shown in the following equation:

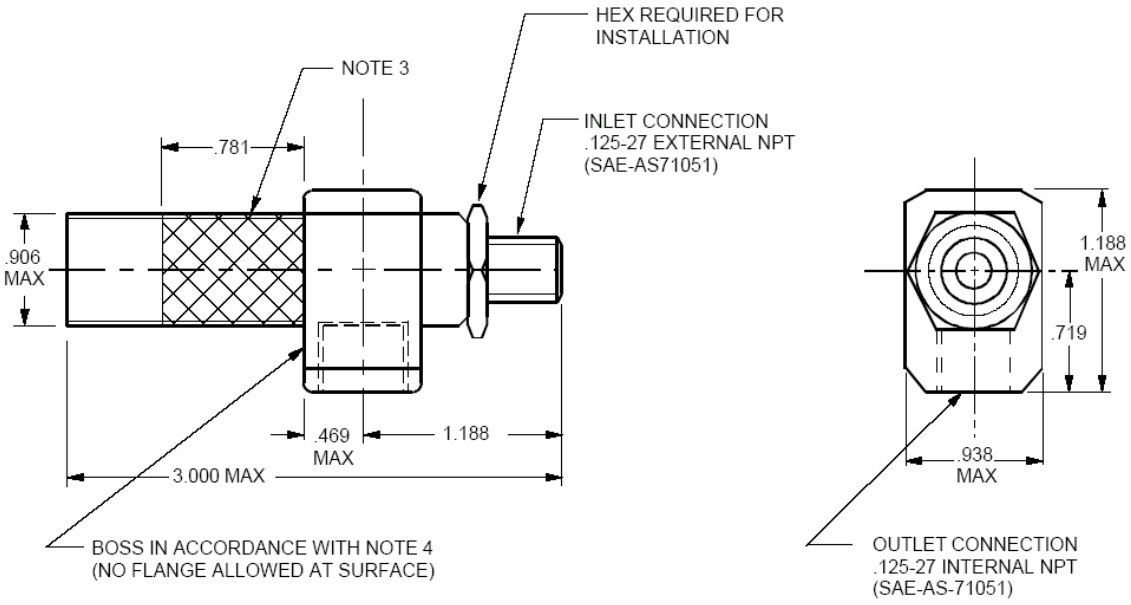
$$\text{Acceleration} = 0.0511(\text{frequency}^2)(\text{displacement})$$

6.10 Subject term (key word) listing.

Breathing
Liquid
LOX converter

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NOTES:

1. Dimensions are in inches.
2. Tolerance is ±.031.
3. Crosshatched surface is for a .875 DIA. mounting clamp.
4. Minimum wall thickness: .063 for forgings and .116 for castings.
5. Figure not to scale.

FIGURE 1. Pressure relief valve

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